Nityashree S

AI Engineer · AI agents, MCP connectors, guardrails and evals

Portfolio & case studies · Personal website
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AI Engineer who builds and secures production AI agents: the platform, the tools they connect to, and the guardrails and evals around them. I direct AI coding agents and verify every change before it ships.

EXPERIENCE

AI Engineer

Jun 2026 – Present

AI Team · Yuno (payment orchestration) · Onsite

PLATFORM AND RELIABILITY

- Root-caused an outage where one malformed instruction file took a production agent offline; fixed it at save and read time with a failing test first, and stopped the error from echoing untrusted text to the model.
- Audited the tool gateway every agent tool call passes through; shipped per-backend timeouts, cancellation isolation, metrics and structured logs. The new logs exposed a connector failing silently in production; I fixed its routing.
- Fixed agent-session failures (orphaned or duplicate tool-call entries that providers rejected; failed turns that lost the user's message) and added a per-turn outcome record to measure agent success in production.

ACCESS AND SECURITY

- Built guest access to personal AI assistants across five services, to rules set by our lead: per-guest permissions, separate credential custody and session isolation; released to production behind a feature switch.
- Built a per-user OAuth (PKCE) connector so each employee's agent can search only what that employee may see; we chose per-user sign-in over a shared key. Live in production.
- Closed web-security gaps (a cross-site request forgery logout, unverified webhook events, unset secrets that let calls through, now fail closed) and patched reachable dependency vulnerabilities across several services.

EVALS AND MODEL BEHAVIOUR

- Built a reply-language resolver that decides which language each agent reply uses; ran it in shadow mode beside the old logic, measured it on a labelled test set, then shipped it to production.
- Showed with a small A/B test, prompt held fixed, that the model, not the prompt, made prebuilt agents ignore instructions; the team changed the default model.

AGENTS FOR BUSINESS TEAMS

- Built finance agents: invoice intake where the model reads and tested code applies the rules, and a mailbox sorter in production whose projected monthly model cost I cut by changing its schedule after measuring an empty run.

Payments Analyst Intern

Dec 2025 – Jun 2026

Office of the CEO · Yuno (payment orchestration) · Onsite

- Built and maintained a payments-capability database of providers, payment methods and markets, and audited it with a documented error taxonomy: duplicates, category mismatches, region errors, unsupported methods.
- Ran acceptance testing for an internal AI assistant that answers payment-coverage questions: collected test sets from Sales and Business Development, and caught a data error before launch.

PROJECTS

- **Smart Router** · Python, FastAPI, PostgreSQL, Redis · Code — An OpenAI-compatible LLM gateway: rules route each prompt to a cheaper or stronger model tier, a failing provider falls back to the next tier, and every request logs its estimated cost. Unit tests run in CI.
- **Meridian** · Python, FastAPI, asyncio · Code — A multi-agent research API: a planner splits the question, researchers search in parallel, embeddings rank passages and a critic flags gaps; each stage streams live.
- **Cross-Border Fee API** · TypeScript, Express, Jest · Code · Live demo — An itemised price estimate for Latin American B2B payments (FX spread, provider fees, local taxes), with rates kept separate from the pricing engine; Jest tests.

SKILLS

AI engineering	LLM agents and tool use, MCP, retrieval and ranking, evals (labelled sets, shadow mode, A/B), guardrails, prompt-injection defence
Security	OAuth 2.0 / PKCE, multi-tenant isolation, webhook signatures, CSRF, fail-closed design, dependency scanning
Shipped to	Go, TypeScript/React, Python · PostgreSQL, MongoDB, Redis · Docker, Kubernetes, AWS · Git, GitHub, CI (via AI coding agents)
How I work	AI-assisted development (Claude Code), failing test first, code review, feature-switched releases

EDUCATION

B.Tech, Computer Science and Engineering (AI & Data Science) 2022 – 2026
SASTRA University

Class XII, CBSE · Convent of Jesus and Mary

CERTIFICATIONS

Supervised Machine Learning
DeepLearning.AI and Stanford University (Coursera)

ML Pipelines with Azure ML Studio
Coursera

TALKS AND RECOMMENDATIONS

Speaker, internal AI talk series · Recommended by Simón Martínez, VP of AI at Yuno: “an exceptionally fast learner”, with work “well architected, thoroughly tested, and production-ready” (letter on request)